Credit Card Fraud Detection Using Machine Learning

Final Comparison Report

1. Project Introduction

During this project, I worked on credit card fraud detection using machine learning techniques. The main purpose of the project was to analyze credit card transaction data and build machine learning models that can help identify fraudulent transactions. I started by understanding the dataset and performing exploratory data analysis. After that, I worked on data preprocessing, checked for missing and duplicate values, analyzed outliers, studied feature correlations, and performed feature scaling. Finally, I trained and compared four machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbours (KNN). The models were compared using accuracy, precision, recall, F1-score, and confusion matrices.

2. Dataset Description

The dataset used for this project is a Credit Card Fraud Detection dataset. It contains transaction information such as transaction date and time, merchant, transaction category, transaction amount, city, state, latitude and longitude, city population, customer job, date of birth, transaction number, merchant latitude and longitude, and Is_Fraud.

The target variable is Is_Fraud. Here, 0 represents a normal/non-fraudulent transaction and 1 represents a fraudulent transaction. The dataset had 15 columns containing both numerical and categorical information.

The final modeling data contained 337,825 non-fraud transactions and 1,782 fraud transactions, showing a strong class imbalance.

3. Algorithms Used

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbours (KNN)

4. Model Performance Comparison

The individual model evaluation results from the notebook are used for the final comparison.

Logistic Regression: Accuracy 99.45%, Precision 0.00%, Recall 0.00%, F1-Score 0.000.

Decision Tree: Accuracy 99.57%, Precision 58.16%, Recall 64.04%, F1-Score 0.610.

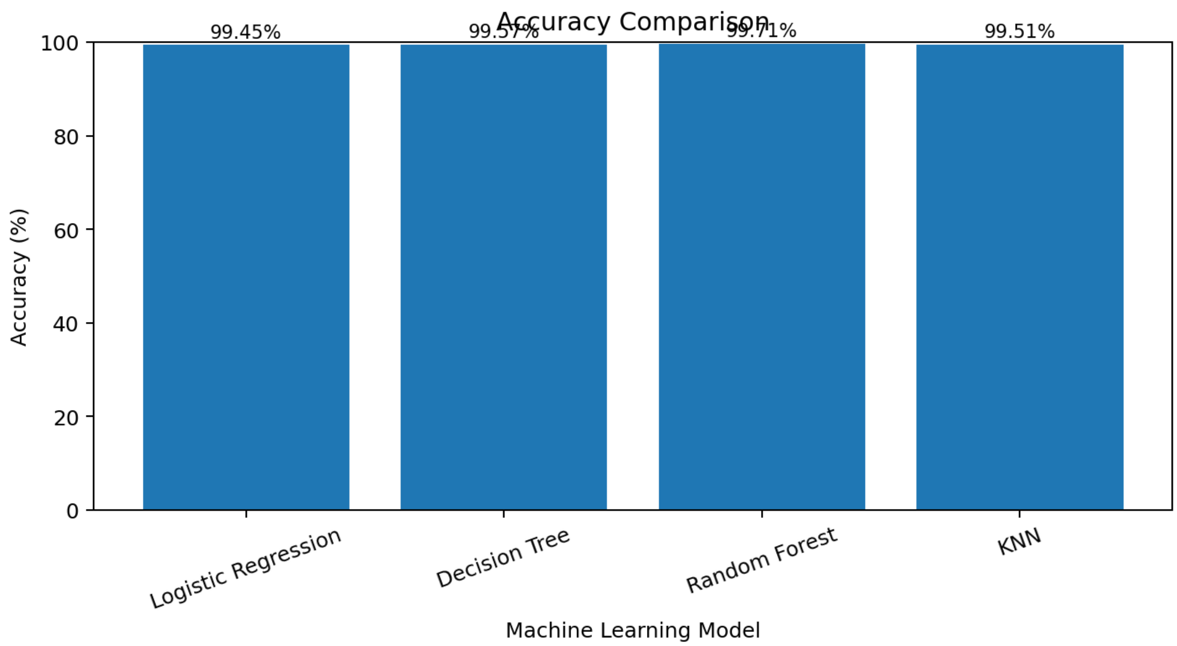
Random Forest: Accuracy 99.71%, Precision 82.72%, Recall 56.46%, F1-Score 0.671.

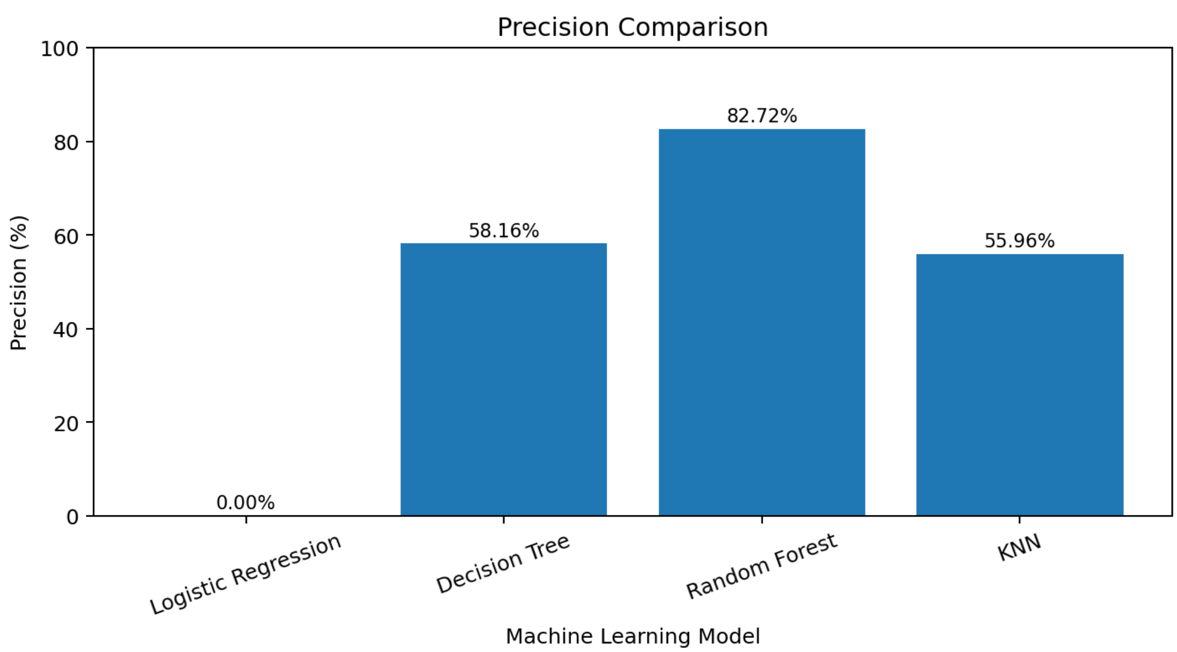
KNN: Accuracy 99.51%, Precision 55.96%, Recall 30.34%, F1-Score 0.393.

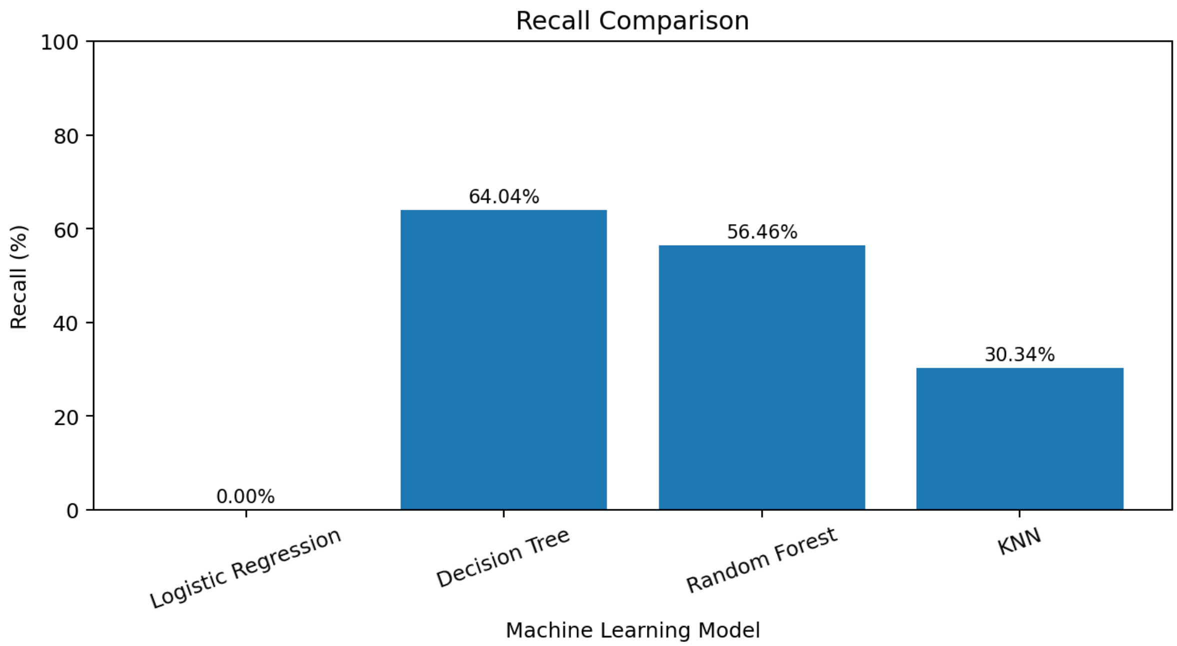
5. Table

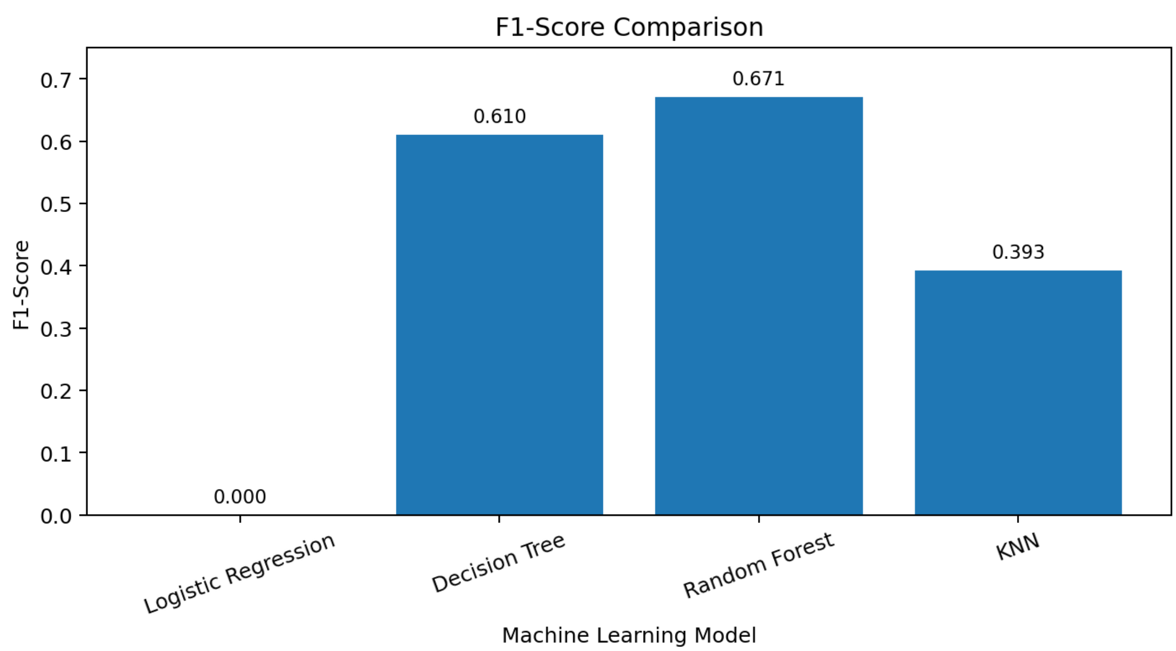
Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	99.45%	0.00%	0.00%	0.000
Decision Tree	99.57%	58.16%	64.04%	0.610
Random Forest	99.71%	82.72%	56.46%	0.671
KNN	99.51%	55.96%	30.34%	0.393

6. Comparison Graphs









The comparison graphs should show Accuracy, Precision, Recall, and F1-Score for Logistic Regression, Decision Tree, Random Forest, and KNN. The graph values are based on the table above.

7. Confusion Matrix Comparison

Logistic Regression: [[67,546, 20], [356, 0]].

Decision Tree: [[67,402, 164], [128, 228]].

Random Forest: [[67,524, 42], [155, 201]].

KNN: [[67,481, 85], [248, 108]].

Logistic Regression correctly identified 0 fraudulent transactions, explaining its 0% fraud recall.

Decision Tree identified 228 of 356 fraud cases and therefore had the highest recall.

Random Forest identified 201 fraud cases and produced only 42 false positives, supporting its high precision.

KNN identified 108 fraud cases.

The sampling experiments increased fraud recall to approximately 71.91%, but also produced many false-positive predictions.

8. Best Model Selection

Metric	Best Model
Accuracy	Random Forest
Precision	Random Forest
Recall	Decision Tree
F1-Score	Random Forest

For the final selection, F1-score was considered because the dataset is highly imbalanced and accuracy alone does not give a good picture of fraud detection performance. Random Forest achieved the highest F1-score at approximately 0.671. Therefore, Random Forest is selected as the preferred overall model.

9. Why this model was selected

I selected Random Forest because it provided the best overall balance between precision and recall among the four tested models. Its accuracy was 99.71%, precision was 82.72%, recall was 56.46%, and F1-score was 67.11%. It achieved the highest accuracy, precision, and F1-score.

Although Decision Tree had a higher recall of 64.04%, Random Forest had much higher precision and the highest F1-score. Therefore, Random Forest is the best overall choice when considering a balance between correctly identifying fraud and avoiding too many false fraud alerts. If maximizing recall were the main priority, Decision Tree could be considered instead.

10. Limitations

- Class Imbalance: The number of normal transactions is much higher than the number of fraudulent transactions, making fraud detection difficult.
- Accuracy: The high accuracy values can be misleading because most transactions are non-fraudulent.
- Fraud Recall: The original models had relatively low recall values, meaning many actual fraudulent transactions were still missed.
- False Positives: The sampling experiments improved recall, but they also resulted in a large number of normal transactions being classified as fraud.
- Feature Correlation: Some geographical features were highly correlated and may contain similar information.
- Model Comparison: Some earlier comparison tables reused the prediction variable, so the individual model evaluation results were used for the final comparison.

11.Final Conclusion

In this project, I worked on credit card fraud detection using machine learning. I started by exploring the transaction data and then performed preprocessing, outlier analysis, correlation analysis, feature scaling, and categorical encoding. I also analyzed the class imbalance problem because fraudulent transactions were much fewer than normal transactions.

I trained four machine learning models: Logistic Regression, Decision Tree, Random Forest, and KNN. From the results, Random Forest achieved the highest accuracy at 99.71%, the highest precision at 82.72%, and the highest F1-score at 0.671. Decision Tree achieved the highest recall at 64.04%.

The sampling experiments showed that balancing the classes can improve fraud recall to approximately 71.91%, but this also creates many false-positive predictions. Therefore, accuracy should not be considered the only important metric in a fraud detection problem. Precision, recall, F1-score, and confusion matrices are also important.

Overall, Random Forest is the best overall model for this project based on the highest F1-score, precision, and accuracy. Decision Tree may be preferred when recall is the main priority, while Random Forest provides the strongest overall balance among the four tested models.